

Niagara Falls? The Pyramids? The Taj Mahal? There, there, and there. It also has GPS coordinates, so if you were to feed them into your GPS receiver, it'll give you directions on how to get there from where you are.

Why it's "evil":

GPS and Google Earth are powerful technologies on their own, but when combined, make for an excellent planning tool. A GPS receiver will give you directions to your destination, and Google Earth shows you what you'll see when you get there. While you're on your way, GPS receivers can't show you visual landmarks, but Google Earth can. And most importantly, when you're on a little boat on its way to Mumbai, and don't really know the way, GPS will let you navigate there easily.

Google Earth has long been criticised by several world leaders — former President A. P. J. Abdul Kalam, among others — for revealing the locations of important military installations across the world. In the US, Google has been asked to obscure important locations such as the White House, and mysteriously, the residence of former vice-president Dick Cheney. The argument was that Google Earth could potentially be, well, a tool in the hands of terrorists.

The Tech: BlackBerrys

What they used it for:

Coordinating with each other, tracking news via GPRS

How it works:

General Packet Radio Service (GPRS) is another technology that's been around for ages, and is relatively simple in concept. It uses the same signal that your voice travels on when you're on a call, only with a few variations to make data transfer faster.

When you're talking with someone (or "exchanging voice information"), all the data that comes and goes from your phone is *circuit-switched* — all the "packets" of data travel in sequence, and on the same frequency. This is important, because you want people to hear you say "hello" and not "lohel". When you're surfing the Internet, however, this isn't as important — you want



your data to reach you as fast as possible, through as many routes as necessary, and your phone will take care of converting all the lohels to hellos. When you request data over your mobile network, your provider sends it to your phones on all the frequencies your phone can access, but not in order. This is called *packet-switching* — each antenna on the network decides the best route for each packet, so each

packet takes the fastest possible route to your phone.

Fighting it

Thankfully, this isn't hard — once the authorities know the offending number, they can shut it down, tap its calls, even track the phone's position. They can also see which sites the phone's user is visiting, and block them if necessary. As for the news, sites could stop reporting live news — especially if that news talks about the police's plans. Then again, stopping live news means that nobody knows what's going on, which means rumours, panic, and all the nastiness that comes with it. As for the e-mail issue, soon after the dispute arose, RIM allowed Indian security agencies to monitor the BlackBerry network, and in July 2008, the service was cleared by the telecom ministry.

Why it's "evil":

It isn't. But you know that already. Having an Internet connection on your phone is about as evil as having a computer at home. Some sites have emphasised the BlackBerry, as if it were a deadly tool in the wrong hands, but really, it's just an innocent device with its name being dragged through the mud. It could just have easily been Nokias or Sony Ericssons in that boat.

There have, however, been concerns regarding the BlackBerry e-mail service. In India, communication technologies are required by law to allow for "lawful interception" — if the government suspects you of unsavoury activities, it needs to be able to monitor your phone, e-mail and other means of communication. The BlackBerry service, however, doesn't allow for such interception, which is why security agencies wanted to ban it. An e-mail service that cannot be monitored could be a great tool in the hands of terrorists, they said.

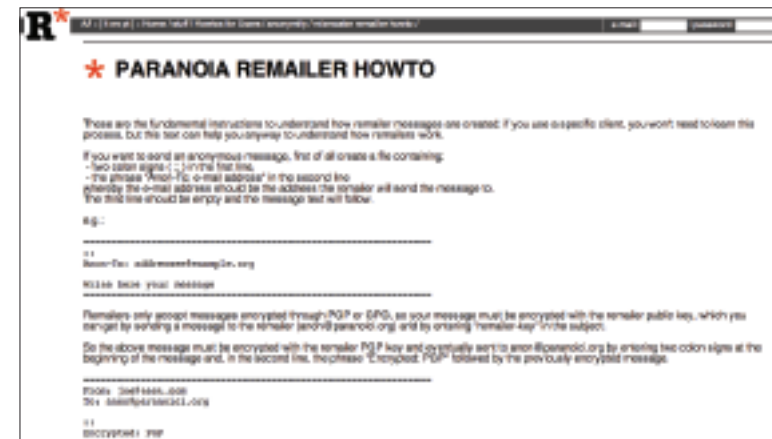
Supposed security risks aside, there is no evidence that suggests that Mumbai terrorists used their BlackBerrys for e-mail.

The Tech: Remailers

What they used it for:

Telling the media that they were responsible, without being traceable.

How it works:



When you don't want the recipient of an e-mail to know where it came from, you first send it to a remailer, which strips the e-mail of all your information, and then sends it on its way, complete with dud reply-to address. They're supposed to make your e-mail anonymous, but there's no such thing as purely anonymous on the Internet — there's just harder to track. And depending on how hard-to-track you want your e-mail to be, you can use one of three types of remailers.

The *pseudonymous remailer* is the simplest — it replaces your e-mail and IP addresses with its own and sends your e-mail. No complex encryp-

Fighting it

Remailers are actually being fought against even as you read this. Unless you have your own remailer server, you never know when one will be taken down. However, sitting back and hoping for remailers to vanish is silly at best. Our forces need better experts, and more importantly, time, if they are to trace e-mails through remailers. So even if the "Deccan Mujahideen" wasn't really responsible for the attacks, tracing their e-mail can lead us to tomorrow's attackers.

What you used

While those three days showed us terrorists can use simple technology for evil, we also saw people attempt use technology for good.

The Tech: Twitter

What you used it for: Telling your friends you were all right, asking people to go donate blood, telling the world what's going on

How it helped: Reassured friends and family, covered news that the agencies weren't reporting

How it didn't: Fostered conspiracy theories, added to the confusion, possibly kept terrorists updated when news channels stopped broadcasting live news

The Tech: Flickr

What you used it for: Documenting the carnage

How it helped: Some photos — especially by Vinukumar Ranganathan — surpassed what the mainstream media could get

How it didn't: Apart from conveying information, pictures aren't much use

tion involved, and the recipient can even reply to the e-mail, which will reach you through the remailer. And of course, because it's so simple, it's easy to tell where an e-mail really came from.

For more secure e-mail, you need a *cypherpunk remailer*, to which you send an encrypted e-mail, which the server then decrypts and sends to the recipient. It's more sophisticated than just hiding your name and IP address, and to make your e-mail even harder to trace, you can have it go through two, three or four remailers. Anyone who'd want to trace your e-mail would have a formidable (though still not impossible) task ahead of them.

And then there's the *mixmaster remailer*, which adds to the goodness of the cypherpunk, and even needs you to have special software installed on your computer. But this isn't entirely anonymous either, but was designed to make it really difficult and time consuming for anyone to trace e-mails.

Why it's "evil":

All right, so this one might qualify as a little evil. People use remailers to cover their tracks, and any activity that involves track-covering can't be good.

The Tech: VoIP phones



What they used it for:

"Phoning home" and getting orders from their handlers

How it works:

Voice over Internet Protocol (VoIP) telephony has been around since the first days of the Internet, and it's simple enough too — it uses your Internet connection to transmit voice between people. The problem with the Internet, however, is that it's a packet-switched network, which means that packets of data don't reach their destination in the right sequence. With voice, of course, this will not do at all — which is why several technologies had to be developed to ensure that voice travelled the pipes without any problems.